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Модуль 1

Практика 2

Раздел 1

```
export MYUSER=$USER
```

```
script "practice2_${MYUSER}_part1.log"
```

```
ssh-keygen -t rsa -b 2048
```

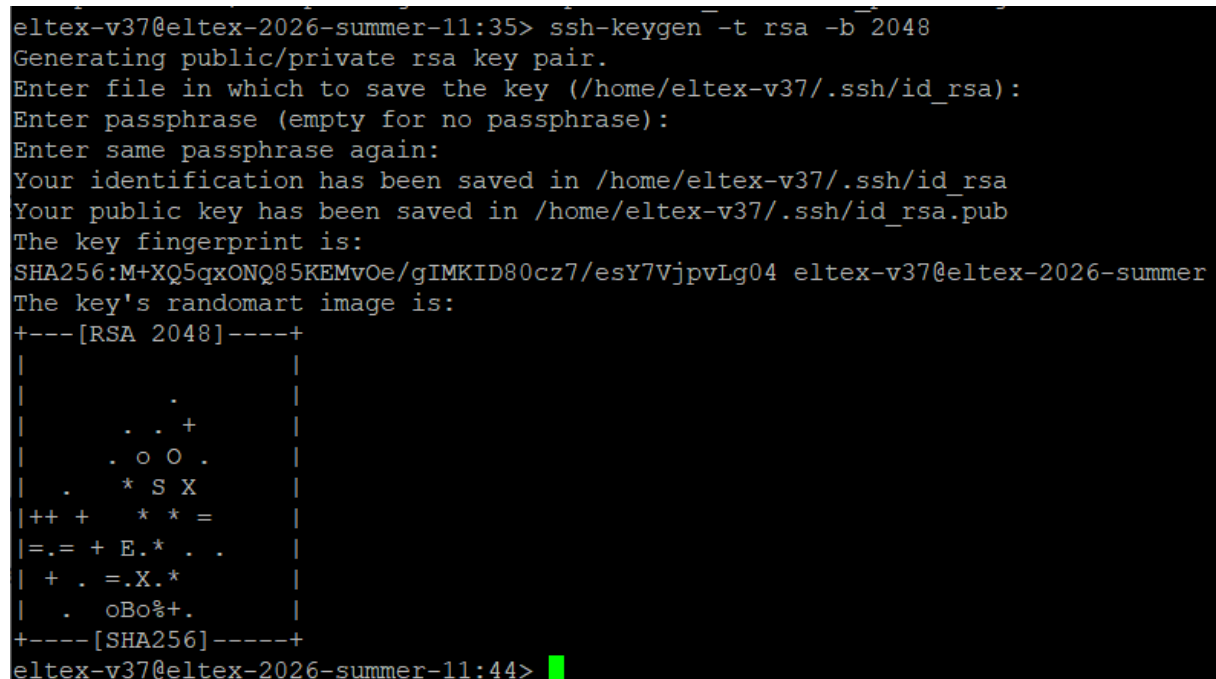
A terminal window showing the execution of the 'ssh-keygen' command. The user is at the prompt 'eltex-v37@eltex-2026-summer-11:35>'. The command 'ssh-keygen -t rsa -b 2048' is entered. The output shows the generation of a public/private RSA key pair. It prompts for a file to save the key (defaulting to '/home/eltex-v37/.ssh/id_rsa'), a passphrase (empty), and confirmation of the passphrase. It then shows the key's fingerprint (SHA256) and a randomart image. The prompt changes to 'eltex-v37@eltex-2026-summer-11:44>'.

Рисунок 1 – Создание ключа ssh

```
ssh-copy-id -i ~/.ssh/id_rsa.pub root@172.16.9.211
```

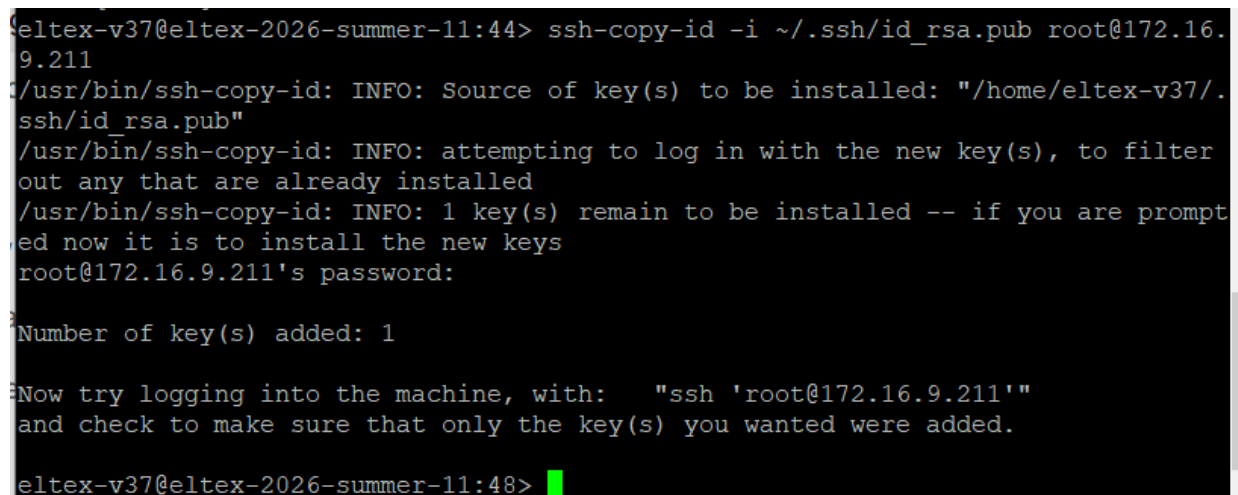
A terminal window showing the execution of the 'ssh-copy-id' command. The user is at the prompt 'eltex-v37@eltex-2026-summer-11:44>'. The command 'ssh-copy-id -i ~/.ssh/id_rsa.pub root@172.16.9.211' is entered. The output shows the source of the key(s) to be installed, an attempt to log in with the new key(s) to filter out any that are already installed, and a message stating that 1 key(s) remain to be installed. It then prompts for the password of 'root@172.16.9.211'. The output shows 'Number of key(s) added: 1' and a message to try logging into the machine with 'ssh 'root@172.16.9.211'' and check to make sure that only the key(s) you wanted were added. The prompt changes to 'eltex-v37@eltex-2026-summer-11:48>'.

Рисунок 2 – Копирование ключа на сервер 2 практики

```
cat ~/.ssh/id_rsa.pub
```

```
ls -l ~/.ssh/id_rsa.pub
```

```
eltex-v37@eltex-2026-summer-11:57> cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQAC7Mrk4fBhXURa6to64ERluip3HrjcAtMyLeBql39ya
BRjgFiFf13N+17hFLVd3OCLW0QV7Y3sPa3sZ7tKOPz+l4yGL84kTEoMxq4O3RCdjzC42Z4yAfHBTbX8Q
eyZseS7+pkEkO2yjLK7BWno4hHILPULtwaKnCuYWJiuQZ7hGSNsgEx0v7gX39Gx5Y/SB9gO2cM7mXh+c
pruuvU8RU3FFYw6NHQRmu6IRtKcyBjteMwTZYaefbQDRY4aJrZOcJDfn+kkesOPFdRjwS040MYCcNbZc
FUF0WlJlOs3IHTR+iMzBbxd1vo9sdVYTvs9hgifPKDULPyg+dC4s+nICCI4R eltex-v37@eltex-202
6-summer
eltex-v37@eltex-2026-summer-11:57> ls -l ~/.ssh/id_rsa.pub
-rw-r--r-- 1 eltex-v37 eltex-v37 409 Jul  8 11:44 /home/eltex-v37/.ssh/id_rsa.pub
eltex-v37@eltex-2026-summer-11:58> █
```

Рисунок 3 – Содержимое ключа и права доступа к файлу ключа

```
ssh root@172.16.9.211
```

```
cat ~/.ssh/authorized_keys
```

```
ls -la ~/.ssh
```

```
root@eltex-practice2-v37:~# cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQAC7Mrk4fBhXURa6to64ERluip3HrjcAtMyLeBql39ya
BRjgFiFf13N+17hFLVd3OCLW0QV7Y3sPa3sZ7tKOPz+l4yGL84kTEoMxq4O3RCdjzC42Z4yAfHBTbX8Q
eyZseS7+pkEkO2yjLK7BWno4hHILPULtwaKnCuYWJiuQZ7hGSNsgEx0v7gX39Gx5Y/SB9gO2cM7mXh+c
pruuvU8RU3FFYw6NHQRmu6IRtKcyBjteMwTZYaefbQDRY4aJrZOcJDfn+kkesOPFdRjwS040MYCcNbZc
FUF0WlJlOs3IHTR+iMzBbxd1vo9sdVYTvs9hgifPKDULPyg+dC4s+nICCI4R eltex-v37@eltex-202
6-summer
root@eltex-practice2-v37:~# ls -la ~/.ssh
total 12
drwx----- 2 root root 4096 Mar 12  2025 .
drwx----- 4 root root 4096 Jul  8 04:30 ..
-rw----- 1 root root  409 Jul  8 04:48 authorized_keys
root@eltex-practice2-v37:~# █
```

Рисунок 4 – Содержимое ключа и права доступа к файлу ключа

```
useradd -m -s /bin/bash user1
```

```
passwd user1
```

```
passwd: u1p
```

```
adduser user2
```

```
passwd: u2p
```

```
adduser user3
```

```
passwd: u3p
```

```

root@eltex-practice2-v37:~# useradd -m -s /bin/bash user1
root@eltex-practice2-v37:~# passwd user1
New password:
Retype new password:
passwd: password updated successfully
root@eltex-practice2-v37:~# adduser user2
info: Adding user `user2' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user2' (1002) ...
info: Adding new user `user2' (1002) with group `user2 (1002)' ...
info: Creating home directory `/home/user2' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user2
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `user2' to supplemental / extra groups `users' ...
info: Adding user `user2' to group `users' ...
root@eltex-practice2-v37:~# adduser user3
info: Adding user `user3' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user3' (1003) ...
info: Adding new user `user3' (1003) with group `user3 (1003)' ...
info: Creating home directory `/home/user3' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user3
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `user3' to supplemental / extra groups `users' ...
info: Adding user `user3' to group `users' ...
root@eltex-practice2-v37:~# █

```

Рисунок 5 – Создание пользователей

```
usermod -s /usr/sbin/nologin user3
```

```
su user3
```

```
su -s /bin/bash user3
```

```
exit
```

```

root@eltex-practice2-v37:~# usermod -s /usr/sbin/nologin user3
root@eltex-practice2-v37:~# su user3
This account is currently not available.
root@eltex-practice2-v37:~# su -s /bin/bash user3
user3@eltex-practice2-v37:/root$ exit
exit
root@eltex-practice2-v37:~# █

```

Рисунок 6 – Изменения пользователя 3

```

groupadd shared_group
usermod -aG shared_group user1
usermod -aG shared_group user2
usermod -aG shared_group user3
id user1
id user2
id user3

```

```

root@eltex-practice2-v37:~# groupadd shared_group
root@eltex-practice2-v37:~# usermod -aG shared_group user1
root@eltex-practice2-v37:~# usermod -aG shared_group user2
root@eltex-practice2-v37:~# usermod -aG shared_group user3
root@eltex-practice2-v37:~# id user1
uid=1001(user1) gid=1001(user1) groups=1001(user1),1004(shared_group)
root@eltex-practice2-v37:~# id user2
uid=1002(user2) gid=1002(user2) groups=1002(user2),100(users),1004(shared_group)
root@eltex-practice2-v37:~# id user3
uid=1003(user3) gid=1003(user3) groups=1003(user3),100(users),1004(shared_group)
root@eltex-practice2-v37:~# █

```

Рисунок 7 – Создание группы

```

mkdir /opt/share
chown :shared_group /opt/share
chmod g+rx,g+s /opt/share
ls -ld /opt/share

```

```

root@eltex-practice2-v37:~# mkdir /opt/share
root@eltex-practice2-v37:~# chown :shared_group /opt/share
root@eltex-practice2-v37:~# chmod g+rx,g+s /opt/share
root@eltex-practice2-v37:~# ls -ld /opt/share
drwxrwsr-x 2 root shared_group 4096 Jul  8 05:29 /opt/share
root@eltex-practice2-v37:~# █

```

Рисунок 8 – Создание и настройка каталога

```
echo "umask 027" >> /home/user1/.bashrc
```

```
su - user1
```

```
cd /opt/share
```

```
touch file_user1
```

```
ls -l file_user1
```

```
su - user2
```

```
cd /opt/share
```

```
touch file_user2
```

```
rm file_user1
```

```
su -s /bin/bash user3
```

```
cd /opt/share
```

```
touch file_user3
```

```
rm file_user2
```

```
su - user1
```

```
cd /opt/share
```

```
rm file_user3
```

```
ls -la /opt/share
```

```

root@eltex-practice2-v37:~# echo "umask 027" >> /home/user1/.bashrc
root@eltex-practice2-v37:~# su - user1
user1@eltex-practice2-v37:~$ umask
0027
user1@eltex-practice2-v37:~$ cd /opt/share
user1@eltex-practice2-v37:/opt/share$ touch file_user1
user1@eltex-practice2-v37:/opt/share$ ls -l file_user1
-rw-r----- 1 user1 shared_group 0 Jul  8 05:49 file_user1
user1@eltex-practice2-v37:/opt/share$
logout
root@eltex-practice2-v37:~# su - user2
user2@eltex-practice2-v37:~$ cd /opt/share
user2@eltex-practice2-v37:/opt/share$ touch file_user2
user2@eltex-practice2-v37:/opt/share$ rm file_user1
rm: remove write-protected regular empty file 'file_user1'? y
user2@eltex-practice2-v37:/opt/share$
logout
root@eltex-practice2-v37:~# su -s /bin/bash user3
user3@eltex-practice2-v37:/root$ cd /opt/share
user3@eltex-practice2-v37:/opt/share$ touch file_user3
user3@eltex-practice2-v37:/opt/share$ rm file_user2
user3@eltex-practice2-v37:/opt/share$
exit
root@eltex-practice2-v37:~# su - user1
user1@eltex-practice2-v37:~$ cd /opt/share
user1@eltex-practice2-v37:/opt/share$ rm file_user3
user1@eltex-practice2-v37:/opt/share$
logout
root@eltex-practice2-v37:~# ls -la /opt/share
total 8
drwxrwsr-x 2 root shared_group 4096 Jul  8 05:55 .
drwxr-xr-x 3 root root          4096 Jul  8 05:29 ..
root@eltex-practice2-v37:~#

```

Рисунок 9 – Удаление файлов

`chmod +t /opt/share`

```

root@eltex-practice2-v37:~# chmod +t /opt/share
root@eltex-practice2-v37:~# ls -ld /opt/share
drwxrwsr-t 2 root shared_group 4096 Jul  8 05:55 /opt/share
root@eltex-practice2-v37:~#

```

Рисунок 10 – Установка sticky bit

`su - user1`

`cd /opt/share`

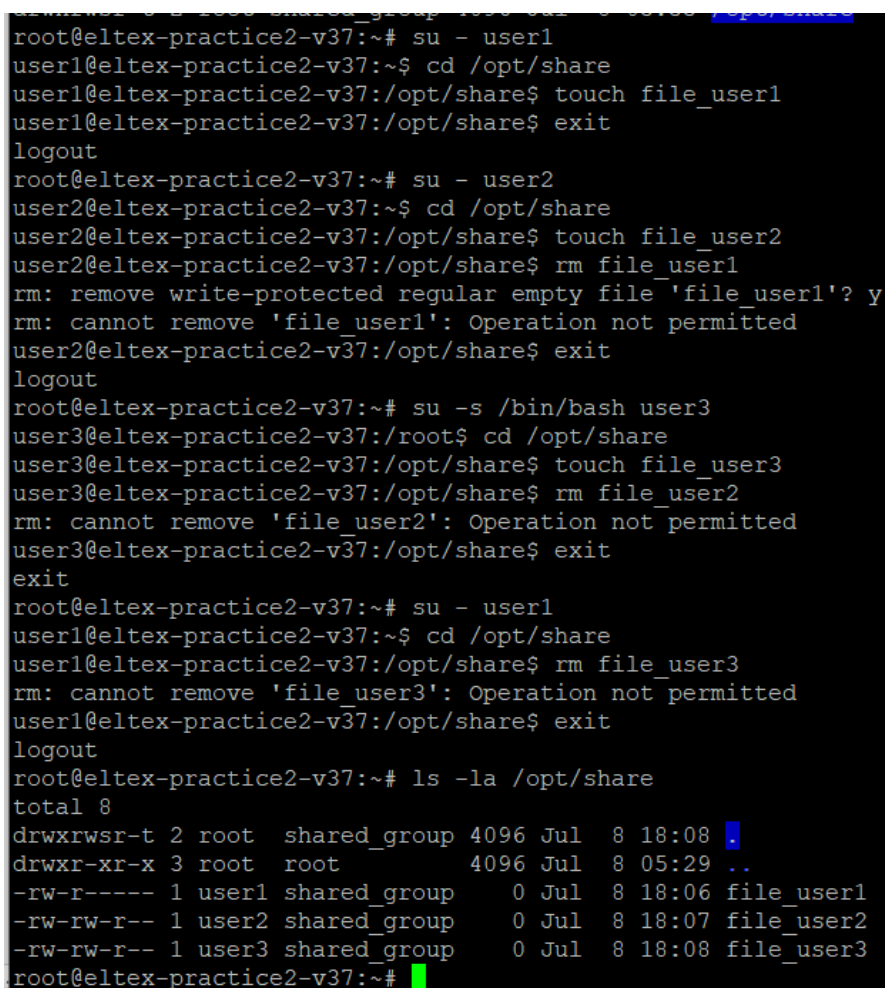
`touch file_user1`

`exit`

`su - user2`

`cd /opt/share`

```
touch file_user2
rm file_user1
exit
su -s /bin/bash user3
cd /opt/share
touch file_user3
rm file_user2
exit
su - user1
cd /opt/share
rm file_user3
exit
ls -la /opt/share
```



```
root@eltex-practice2-v37:~# su - user1
user1@eltex-practice2-v37:~$ cd /opt/share
user1@eltex-practice2-v37:/opt/share$ touch file_user1
user1@eltex-practice2-v37:/opt/share$ exit
logout
root@eltex-practice2-v37:~# su - user2
user2@eltex-practice2-v37:~$ cd /opt/share
user2@eltex-practice2-v37:/opt/share$ touch file_user2
user2@eltex-practice2-v37:/opt/share$ rm file_user1
rm: remove write-protected regular empty file 'file_user1'? y
rm: cannot remove 'file_user1': Operation not permitted
user2@eltex-practice2-v37:/opt/share$ exit
logout
root@eltex-practice2-v37:~# su -s /bin/bash user3
user3@eltex-practice2-v37:/root$ cd /opt/share
user3@eltex-practice2-v37:/opt/share$ touch file_user3
user3@eltex-practice2-v37:/opt/share$ rm file_user2
rm: cannot remove 'file_user2': Operation not permitted
user3@eltex-practice2-v37:/opt/share$ exit
exit
root@eltex-practice2-v37:~# su - user1
user1@eltex-practice2-v37:~$ cd /opt/share
user1@eltex-practice2-v37:/opt/share$ rm file_user3
rm: cannot remove 'file_user3': Operation not permitted
user1@eltex-practice2-v37:/opt/share$ exit
logout
root@eltex-practice2-v37:~# ls -la /opt/share
total 8
drwxrwsr-t 2 root  shared_group 4096 Jul  8 18:08 .
drwxr-xr-x 3 root  root          4096 Jul  8 05:29 ..
-rw-r----- 1 user1 shared_group    0 Jul  8 18:06 file_user1
-rw-rw-r-- 1 user2 shared_group    0 Jul  8 18:07 file_user2
-rw-rw-r-- 1 user3 shared_group    0 Jul  8 18:08 file_user3
root@eltex-practice2-v37:~#
```

Рисунок 11 – Удаление файлов

which dmesg

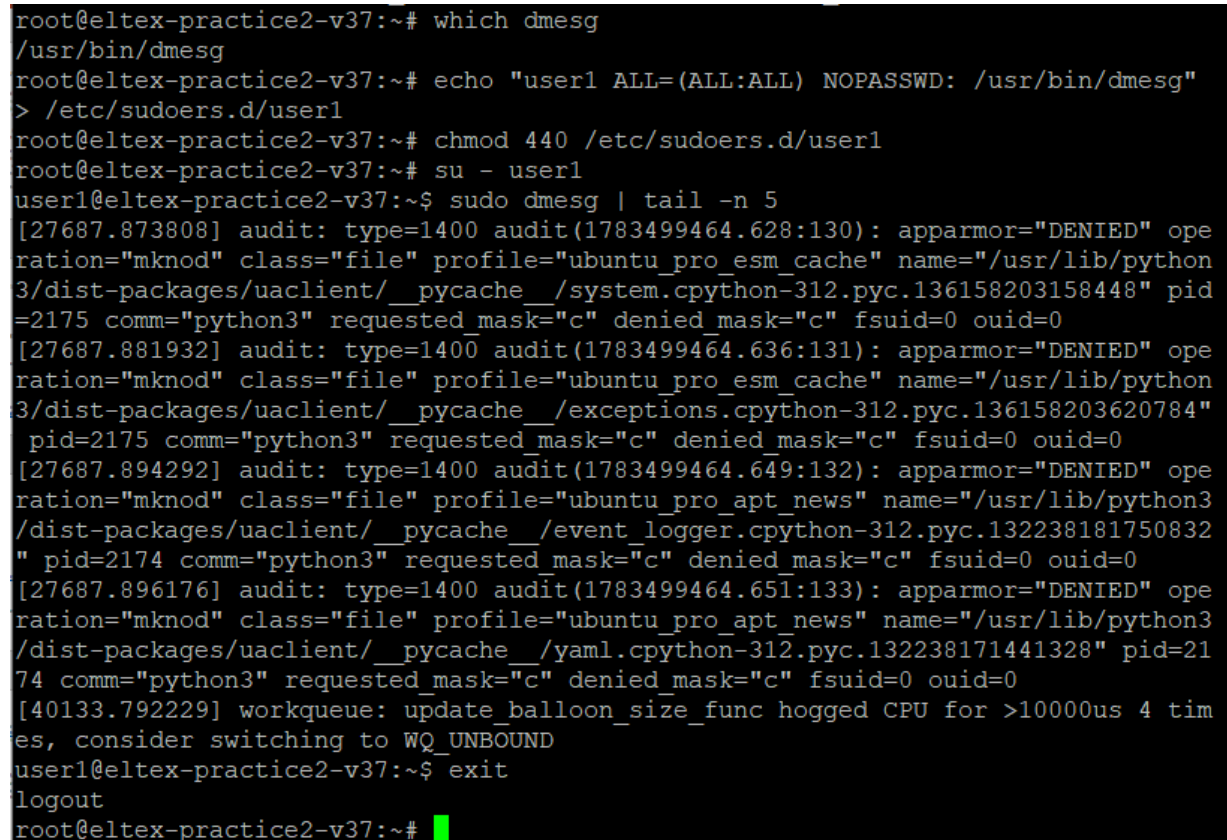
```
echo "user1 ALL=(ALL:ALL) NOPASSWD: /usr/bin/dmesg" >
/etc/sudoers.d/user1
```

```
chmod 440 /etc/sudoers.d/user1
```

```
su - user1
```

```
sudo dmesg | tail -n 5
```

```
exit
```



```
root@eltex-practice2-v37:~# which dmesg
/usr/bin/dmesg
root@eltex-practice2-v37:~# echo "user1 ALL=(ALL:ALL) NOPASSWD: /usr/bin/dmesg"
> /etc/sudoers.d/user1
root@eltex-practice2-v37:~# chmod 440 /etc/sudoers.d/user1
root@eltex-practice2-v37:~# su - user1
user1@eltex-practice2-v37:~$ sudo dmesg | tail -n 5
[27687.873808] audit: type=1400 audit(1783499464.628:130): apparmor="DENIED" ope
ration="mknod" class="file" profile="ubuntu_pro_esm_cache" name="/usr/lib/python
3/dist-packages/uaclient/__pycache__/system.cpython-312.pyc.136158203158448" pid
=2175 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[27687.881932] audit: type=1400 audit(1783499464.636:131): apparmor="DENIED" ope
ration="mknod" class="file" profile="ubuntu_pro_esm_cache" name="/usr/lib/python
3/dist-packages/uaclient/__pycache__/exceptions.cpython-312.pyc.136158203620784"
pid=2175 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[27687.894292] audit: type=1400 audit(1783499464.649:132): apparmor="DENIED" ope
ration="mknod" class="file" profile="ubuntu_pro_apt_news" name="/usr/lib/python3
/dist-packages/uaclient/__pycache__/event_logger.cpython-312.pyc.132238181750832
" pid=2174 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[27687.896176] audit: type=1400 audit(1783499464.651:133): apparmor="DENIED" ope
ration="mknod" class="file" profile="ubuntu_pro_apt_news" name="/usr/lib/python3
/dist-packages/uaclient/__pycache__/yaml.cpython-312.pyc.132238171441328" pid=21
74 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[40133.792229] workqueue: update_balloon_size_func hogged CPU for >10000us 4 tim
es, consider switching to WQ_UNBOUND
user1@eltex-practice2-v37:~$ exit
logout
root@eltex-practice2-v37:~#
```

Рисунок 12 – Разрешение user1 выполнять привилегированную команду dmesg

```
nano /opt/run_dmesg.sh
```

```
chown root:root /opt/run_dmesg.sh
```

```
chmod 4755 /opt/run_dmesg.sh
```

```
su - user2
```

```
/opt/run_dmesg.sh
```

```
exit
```

```
nano /opt/run_dmesg.c
```

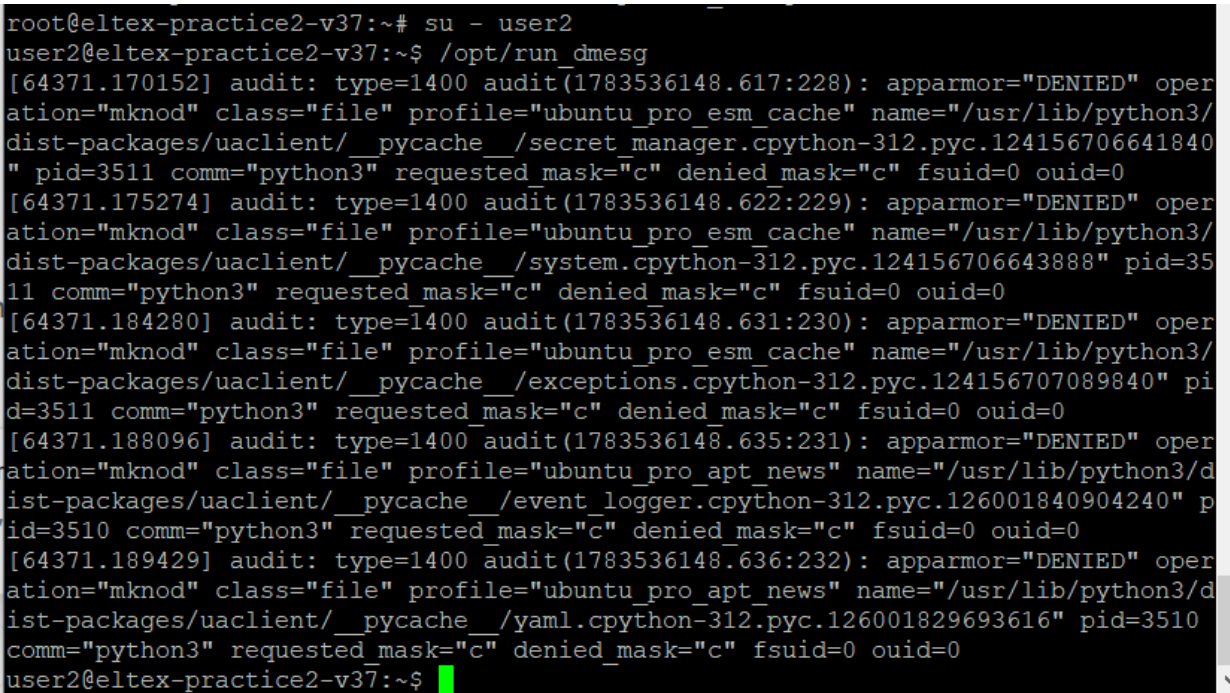
```
#include <stdlib.h>
```



```
#include <unistd.h>

int main() {
    setuid(0);
    system("/usr/bin/dmesg | tail -n 5");
    return 0;
}

gcc /opt/run_dmesg.c -o /opt/run_dmesg
apt update && apt install gcc -y
gcc /opt/run_dmesg.c -o /opt/run_dmesg
chown root:root /opt/run_dmesg
chmod 4755 /opt/run_dmesg
su - user2
/opt/run_dmesg
```



```
root@eltex-practice2-v37:~# su - user2
user2@eltex-practice2-v37:~$ /opt/run_dmesg
[64371.170152] audit: type=1400 audit(1783536148.617:228): apparmor="DENIED" operation="mknod" class="file" profile="ubuntu_pro_esm_cache" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/secret_manager.cpython-312.pyc.124156706641840" pid=3511 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[64371.175274] audit: type=1400 audit(1783536148.622:229): apparmor="DENIED" operation="mknod" class="file" profile="ubuntu_pro_esm_cache" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/system.cpython-312.pyc.124156706643888" pid=3511 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[64371.184280] audit: type=1400 audit(1783536148.631:230): apparmor="DENIED" operation="mknod" class="file" profile="ubuntu_pro_esm_cache" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/exceptions.cpython-312.pyc.124156707089840" pid=3511 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[64371.188096] audit: type=1400 audit(1783536148.635:231): apparmor="DENIED" operation="mknod" class="file" profile="ubuntu_pro_apr_news" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/event_logger.cpython-312.pyc.126001840904240" pid=3510 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[64371.189429] audit: type=1400 audit(1783536148.636:232): apparmor="DENIED" operation="mknod" class="file" profile="ubuntu_pro_apr_news" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/yaml.cpython-312.pyc.126001829693616" pid=3510 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
user2@eltex-practice2-v37:~$
```

Рисунок 13 – Разрешение user2 выполнять привилегированную команду dmesg

```
chage -M 10 user1
chage -M 10 user2
chage -M 10 user3
chage -l user1
```

chage -l user2

chage -l user3

```
root@eltex-practice2-v37:~# chage -M 10 user1
root@eltex-practice2-v37:~# chage -M 10 user2
root@eltex-practice2-v37:~# chage -M 10 user3
root@eltex-practice2-v37:~# chage -l user1
Last password change           : Jul 08, 2026
Password expires               : Jul 18, 2026
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 0
Maximum number of days between password change : 10
Number of days of warning before password expires : 7
root@eltex-practice2-v37:~# chage -l user2
Last password change           : Jul 08, 2026
Password expires               : Jul 18, 2026
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 0
Maximum number of days between password change : 10
Number of days of warning before password expires : 7
root@eltex-practice2-v37:~# chage -l user3
Last password change           : Jul 08, 2026
Password expires               : Jul 18, 2026
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 0
Maximum number of days between password change : 10
Number of days of warning before password expires : 7
root@eltex-practice2-v37:~#
```

Рисунок 14 – Время действия пароля

nano /etc/motd

cat /etc/motd

```
root@eltex-practice2-v37:~# nano /etc/motd
root@eltex-practice2-v37:~# cat /etc/motd
Виктория Царенкова
root@eltex-practice2-v37:~#
```

Рисунок 15 – Редактирование файла /etc/motd

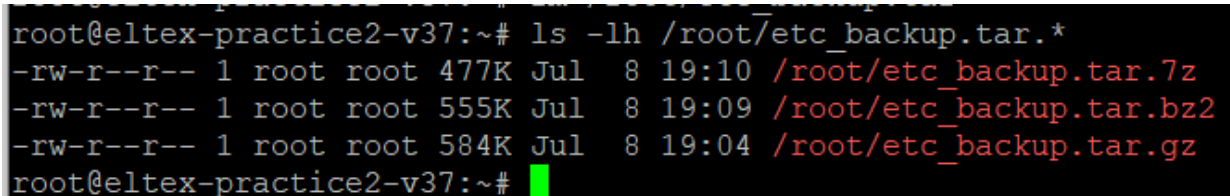
rsync -av /etc/ /root/etc_backup/

```
xdg/autostart/
xdg/autostart/snap-userd-autostart.desktop
xdg/autostart/xdg-user-dirs.desktop
xdg/systemd/
xdg/systemd/user -> ../../systemd/user
xml/
xml/catalog
xml/catalog.old
xml/polkitd.xml
xml/polkitd.xml.old
xml/xml-core.xml
xml/xml-core.xml.old

sent 2,451,825 bytes  received 19,435 bytes  1,647,506.67 bytes/sec
total size is 2,367,426  speedup is 0.96
root@eltex-practice2-v37:~#
```

Рисунок 16 – Бекап

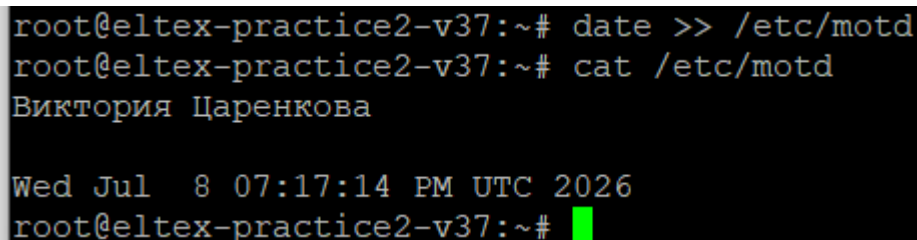
```
apt update && apt install p7zip-full -y
tar -czf /root/etc_backup.tar.gz -C /root etc_backup
tar -cjf /root/etc_backup.tar.bz2 -C /root etc_backup
apt update && apt install bzip2 -y
tar -cjf /root/etc_backup.tar.bz2 -C /root etc_backup
tar -cf /root/etc_backup.tar -C /root etc_backup
7z a /root/etc_backup.tar.7z /root/etc_backup.tar
rm /root/etc_backup.tar
ls -lh /root/etc_backup.tar.*
```



```
root@eltex-practice2-v37:~# ls -lh /root/etc_backup.tar.*
-rw-r--r-- 1 root root 477K Jul  8 19:10 /root/etc_backup.tar.7z
-rw-r--r-- 1 root root 555K Jul  8 19:09 /root/etc_backup.tar.bz2
-rw-r--r-- 1 root root 584K Jul  8 19:04 /root/etc_backup.tar.gz
root@eltex-practice2-v37:~#
```

Рисунок 17 – Размеры полученных файлов

```
date >> /etc/motd
cat /etc/motd
```

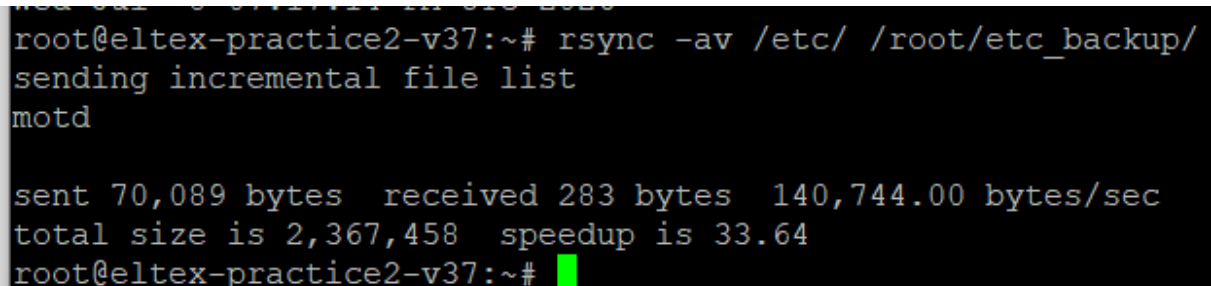


```
root@eltex-practice2-v37:~# date >> /etc/motd
root@eltex-practice2-v37:~# cat /etc/motd
Виктория Царенкова

Wed Jul  8 07:17:14 PM UTC 2026
root@eltex-practice2-v37:~#
```

Рисунок 18 – Добавление даты и времени

```
rsync -av /etc/ /root/etc_backup/
```



```
root@eltex-practice2-v37:~# rsync -av /etc/ /root/etc_backup/
sending incremental file list
motd

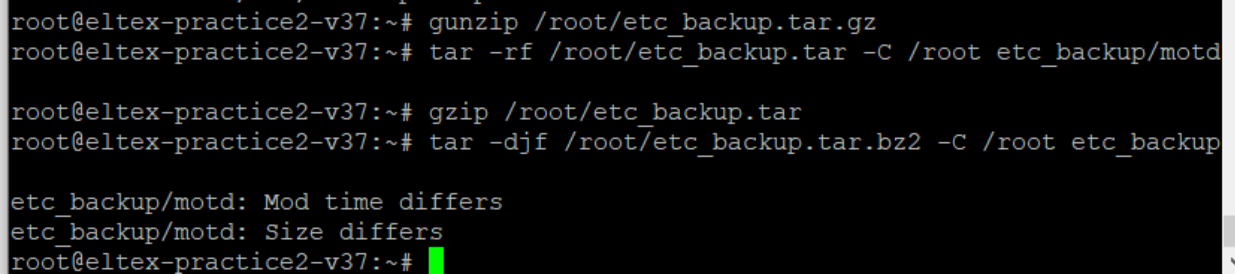
sent 70,089 bytes  received 283 bytes  140,744.00 bytes/sec
total size is 2,367,458  speedup is 33.64
root@eltex-practice2-v37:~#
```

Рисунок 19 – Синхронизация с бекапом

```
gunzip /root/etc_backup.tar.gz
tar -rf /root/etc_backup.tar -C /root etc_backup/motd
```

gzip /root/etc_backup.tar

tar -djf /root/etc_backup.tar.bz2 -C /root etc_backup



```
root@eltex-practice2-v37:~# gunzip /root/etc_backup.tar.gz
root@eltex-practice2-v37:~# tar -rf /root/etc_backup.tar -C /root etc_backup/motd

root@eltex-practice2-v37:~# gzip /root/etc_backup.tar
root@eltex-practice2-v37:~# tar -djf /root/etc_backup.tar.bz2 -C /root etc_backup

etc_backup/motd: Mod time differs
etc_backup/motd: Size differs
root@eltex-practice2-v37:~#
```

Рисунок 20 – Сравнение содержимое архива, упакованного bzip2 с сорержжимым каталога /root/etc_backup

mkdir /root/etc_backup_gzip

mkdir /root/etc_backup_7zip

tar -xzf /root/etc_backup.tar.gz -C /root/etc_backup_gzip

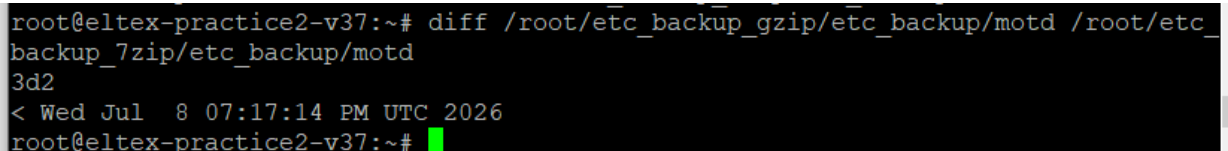
7z x /root/etc_backup.tar.7z -o/root/etc_backup_7zip

tar -xf /root/etc_backup_7zip/etc_backup.tar -C /root/etc_backup_7zip

rm /root/etc_backup_7zip/etc_backup.tar

diff /root/etc_backup_gzip/etc_backup/motd

/root/etc_backup_7zip/etc_backup/motd



```
root@eltex-practice2-v37:~# diff /root/etc_backup_gzip/etc_backup/motd /root/etc_
backup_7zip/etc_backup/motd
3d2
< Wed Jul  8 07:17:14 PM UTC 2026
root@eltex-practice2-v37:~#
```

Рисунок 20 – Сравнение программой diff файлы motd

col -b < practice2_eltex-v37_part1.log > temp_clean.log && mv temp_clean.log
practice2_eltex-v37_part1.log

sed -i -E "s/\x1B\[([0-9;])*[a-zA-Z]//g; s/\??2004[hl]//g; s/3mexp//g; s/3m//g;
s/23m//g; s/<8>//g; s/echo3/echo /g; s/exportR/export /g" practice2_eltex-
v37_part1.log

Раздел 2

```
export MYUSER=$USER
```

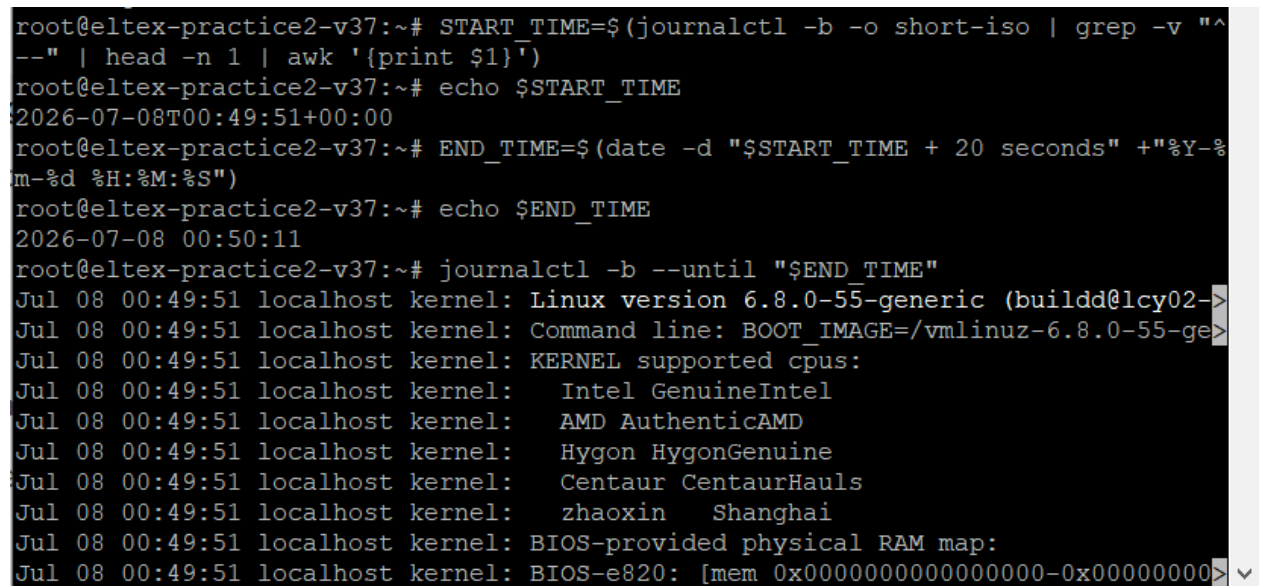
```
script "practice2_${MYUSER}_part2.log"
```

```
ssh root@172.16.9.211
```

```
START_TIME=$(journalctl -b -o short-iso | grep -v "^--" | head -n 1 | awk '{print $1}')
```

```
END_TIME=$(date -d "$START_TIME + 20 seconds" +"%Y-%m-%d %H:%M:%S")
```

```
journalctl -b --until "$END_TIME"
```



```
root@eltex-practice2-v37:~# START_TIME=$(journalctl -b -o short-iso | grep -v "^--" | head -n 1 | awk '{print $1}')
root@eltex-practice2-v37:~# echo $START_TIME
2026-07-08T00:49:51+00:00
root@eltex-practice2-v37:~# END_TIME=$(date -d "$START_TIME + 20 seconds" +"%Y-%m-%d %H:%M:%S")
root@eltex-practice2-v37:~# echo $END_TIME
2026-07-08 00:50:11
root@eltex-practice2-v37:~# journalctl -b --until "$END_TIME"
Jul 08 00:49:51 localhost kernel: Linux version 6.8.0-55-generic (builddd@lcy02-
Jul 08 00:49:51 localhost kernel: Command line: BOOT_IMAGE=/vmlinuz-6.8.0-55-ge
Jul 08 00:49:51 localhost kernel: KERNEL supported cpus:
Jul 08 00:49:51 localhost kernel: Intel GenuineIntel
Jul 08 00:49:51 localhost kernel: AMD AuthenticAMD
Jul 08 00:49:51 localhost kernel: Hygon HygonGenuine
Jul 08 00:49:51 localhost kernel: Centaur CentaurHauls
Jul 08 00:49:51 localhost kernel: zhaoxin Shanghai
Jul 08 00:49:51 localhost kernel: BIOS-provided physical RAM map:
Jul 08 00:49:51 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000
```

Рисунок 21 – Записи из лога загрузки

```
find /var/log -name "auth.log"
```

```
awk '{src=$3; $1=$2=$3=""; print src, $0}' /var/log/auth.log | sed -E 's/^[[0-9]]+\]/g' | sort
```

```
awk '{print $3}' /var/log/auth.log | sed -E 's/^[[0-9]]+\]:?//g' | sort | uniq -c | sort -rn
```

```

useradd: add 'tech' to shadow group 'sudo'
useradd: new group: name=tech, GID=1000
useradd: new group: name=user1, GID=1001
useradd: new user: name=tech, UID=1000, GID=1000, home=/home/tech, shell=/bin
/bash, from=none
useradd: new user: name=user1, UID=1001, GID=1001, home=/home/user1, shell=/b
in/bash, from=/dev/pts/0
useradd: new user: name=user2, UID=1002, GID=1002, home=/home/user2, shell=/b
in/bash, from=/dev/pts/0
useradd: new user: name=user3, UID=1003, GID=1003, home=/home/user3, shell=/b
in/bash, from=/dev/pts/0
usermod: add 'user1' to group 'shared_group'
usermod: add 'user1' to shadow group 'shared_group'
usermod: add 'user2' to group 'shared_group'
usermod: add 'user2' to shadow group 'shared_group'
usermod: add 'user3' to group 'shared_group'
usermod: add 'user3' to shadow group 'shared_group'
usermod: change user 'user3' shell from '/bin/bash' to '/usr/sbin/nologin'
root@eltex-practice2-v37:~#

```

Рисунок 22 – Источники и их сообщения

```

root@eltex-practice2-v37:~# awk '{print $3}' /var/log/auth.log | sed -E 's/\[[0-
9]+\]:?//g' | sort | uniq -c | sort -rn
    284 CRON
     42 su
     38 systemd-logind
     37 sshd
     36 polkitd
     18 useradd
      9 groupadd
      8 (systemd):
      7 usermod
      6 sudo:
      5 passwd
      3 login
      3 chage
      2 gpasswd
      2 chfn
      1 nologin:
      1 chpasswd
root@eltex-practice2-v37:~#

```

Рисунок 23 – Количество повторений

```

awk -F: '$7 == "/usr/sbin/nologin" {print $3, $1}' /etc/passwd | sort -k1 -rn | while
read uid user; do groups=$(groups "$user" | cut -d: -f2- | sed 's/^ //; s/ /,/g'); echo
"$uid, $user, [$groups]"; done

```

```

root@eltex-practice2-v37:~# awk -F: '$7 == "/usr/sbin/nologin" {print $3, $1}' /
etc/passwd | sort -k1 -rn | while read uid user; do groups=$(groups "$user" | cu
t -d: -f2- | sed 's/^ //; s/ /,/g'); echo "$uid, $user, [$groups]"; done
65534, nobody, [nogroup]
1003, user3, [user3,users,shared_group]
998, systemd-network, [systemd-network]
997, systemd-timesync, [systemd-timesync]
992, systemd-resolve, [systemd-resolve]
991, polkitd, [polkitd]
989, fwupd-refresh, [fwupd-refresh]
109, sshd, [nogroup]
108, usbmux, [plugdev]
107, landscape, [landscape]
105, tcpdump, [tcpdump]
104, uidd, [uidd]
103, syslog, [syslog,adm]
101, messagebus, [messagebus]
42, _apt, [nogroup]
39, irc, [irc]
38, list, [list]
34, backup, [backup]
33, www-data, [www-data]
13, proxy, [proxy]
10, uucp, [uucp]
9, news, [news]
8, mail, [mail]
7, lp, [lp]
6, man, [man]
5, games, [games]
3, sys, [sys]
2, bin, [bin]
1, daemon, [daemon]
root@eltex-practice2-v37:~#

```

Рисунок 24 – nologin

dmesg | grep "kernel"

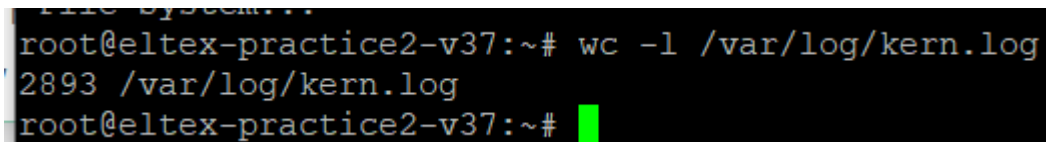
```

root@eltex-practice2-v37:~# dmesg | grep "kernel"
[ 0.157853] Booting paravirtualized kernel on KVM
[ 0.159433] Unknown kernel command line parameters "BOOT_IMAGE=/vmlinuz-6.8.0-5
5-generic", will be passed to user space.
[ 0.335598] Memory: 3933488K/4193760K available (22528K kernel code, 4443K rwd
ta, 14344K rodata, 4988K init, 4716K bss, 260012K reserved, 0K cma-reserved)
[ 1.738319] Loaded X.509 cert 'Build time autogenerated kernel key: e0bedf6fa04
9a1e09bd65ebab3904f14cdd5bc68'
[ 1.997763] Loaded X.509 cert 'Build time autogenerated kernel key: e0bedf6fa04
9a1e09bd65ebab3904f14cdd5bc68'
[ 2.045125] Freeing unused kernel image (initmem) memory: 4988K
[ 2.047424] Write protecting the kernel read-only data: 38912k
[ 2.052851] Freeing unused kernel image (rodata/data gap) memory: 2040K
[ 15.074776] systemd[1]: Listening on systemd-udevd-kernel.socket - udev Kernel
Socket.
[ 15.116247] systemd[1]: Mounting sys-kernel-debug.mount - Kernel Debug File Sys
tem...
[ 15.119158] systemd[1]: Mounting sys-kernel-tracing.mount - Kernel Trace File S
ystem...
[ 15.426573] systemd[1]: Mounted sys-kernel-debug.mount - Kernel Debug File Syst
em.
[ 15.430330] systemd[1]: Mounted sys-kernel-tracing.mount - Kernel Trace File Sy
stem.
[ 15.492989] systemd[1]: Mounting sys-kernel-config.mount - Kernel Configuration
File System...
root@eltex-practice2-v37:~#

```

Рисунок 25 – kernel

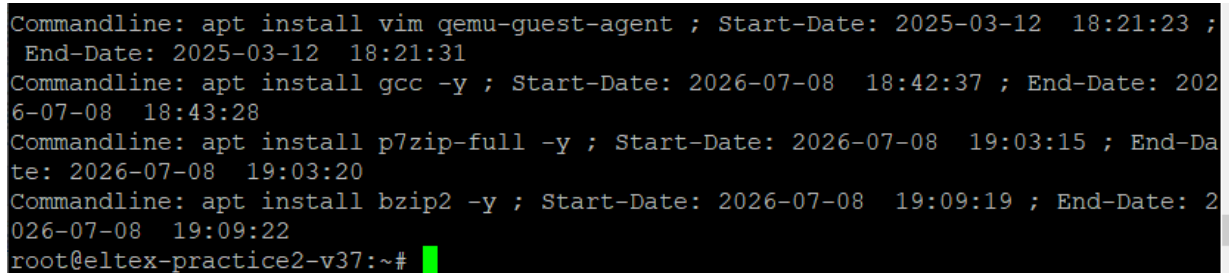

```
wc -l /var/log/kern.log
```



```
root@eltex-practice2-v37:~# wc -l /var/log/kern.log
2893 /var/log/kern.log
root@eltex-practice2-v37:~#
```

Рисунок 26 – Количество строк в файле /var/log/kern.log

```
awk 'BEGIN{RS=""; FS="\n"} {cmd=""; start=""; end=""; for(i=1; i<=NF; i++){
if($i ~ /^Commandline:/) cmd=$i; if($i ~ /^Start-Date:/) start=$i; if($i ~ /^End-
Date:/) end=$i } if(cmd != "") print cmd " ; " start " ; " end}'
/var/log/apt/history.log
```



```
Commandline: apt install vim qemu-guest-agent ; Start-Date: 2025-03-12 18:21:23 ;
End-Date: 2025-03-12 18:21:31
Commandline: apt install gcc -y ; Start-Date: 2026-07-08 18:42:37 ; End-Date: 202
6-07-08 18:43:28
Commandline: apt install p7zip-full -y ; Start-Date: 2026-07-08 19:03:15 ; End-Da
te: 2026-07-08 19:03:20
Commandline: apt install bzip2 -y ; Start-Date: 2026-07-08 19:09:19 ; End-Date: 2
026-07-08 19:09:22
root@eltex-practice2-v37:~#
```

Рисунок 27 – Форматирование вывода записей

```
col -b < practice2_eltex-v37_part2.log > temp_clean.log && mv temp_clean.log
practice2_eltex-v37_part2.log
```

```
sed -i -E "s/\x1B\[[0-9;?]*[a-zA-Z]//g; s/3mexp//g; s/3m//g; s/23m//g; s/<8>//g;
s/echo3/echo /g; s/exportR/export /g" practice2_eltex-v37_part2.log
```

Раздел 3

```
export MYUSER=$USER
```

```
script "practice2_${MYUSER}_part3.log"
```

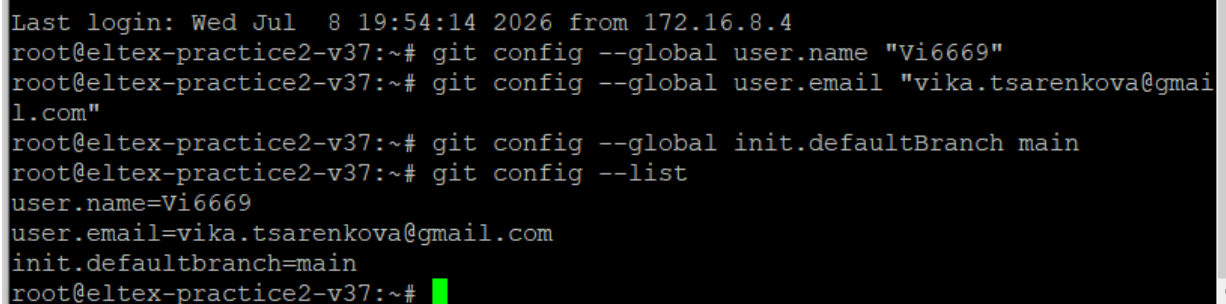
```
ssh root@172.16.9.211
```

```
git config --global user.name "Vi6669"
```

```
git config --global user.email "vika.tsarenkova@gmail.com"
```

```
git config --global init.defaultBranch main
```

```
git config --list
```



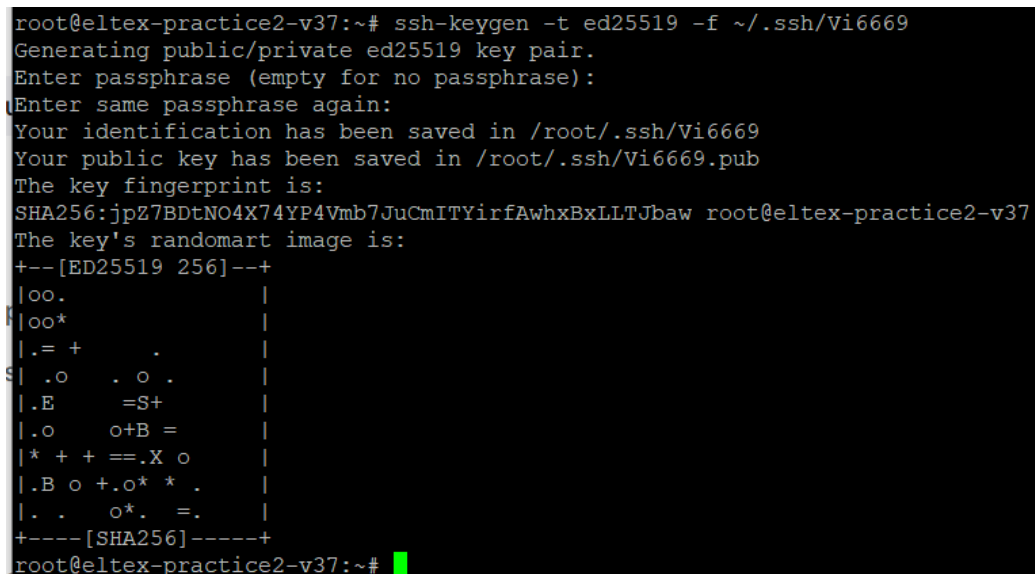
```
Last login: Wed Jul  8 19:54:14 2026 from 172.16.8.4
root@eltex-practice2-v37:~# git config --global user.name "Vi6669"
root@eltex-practice2-v37:~# git config --global user.email "vika.tsarenkova@gmail.com"
root@eltex-practice2-v37:~# git config --global init.defaultBranch main
root@eltex-practice2-v37:~# git config --list
user.name=Vi6669
user.email=vika.tsarenkova@gmail.com
init.defaultbranch=main
root@eltex-practice2-v37:~#
```

Рисунок 28 – Настройка гитхаба

```
apt update
```

```
apt install git gh -y
```

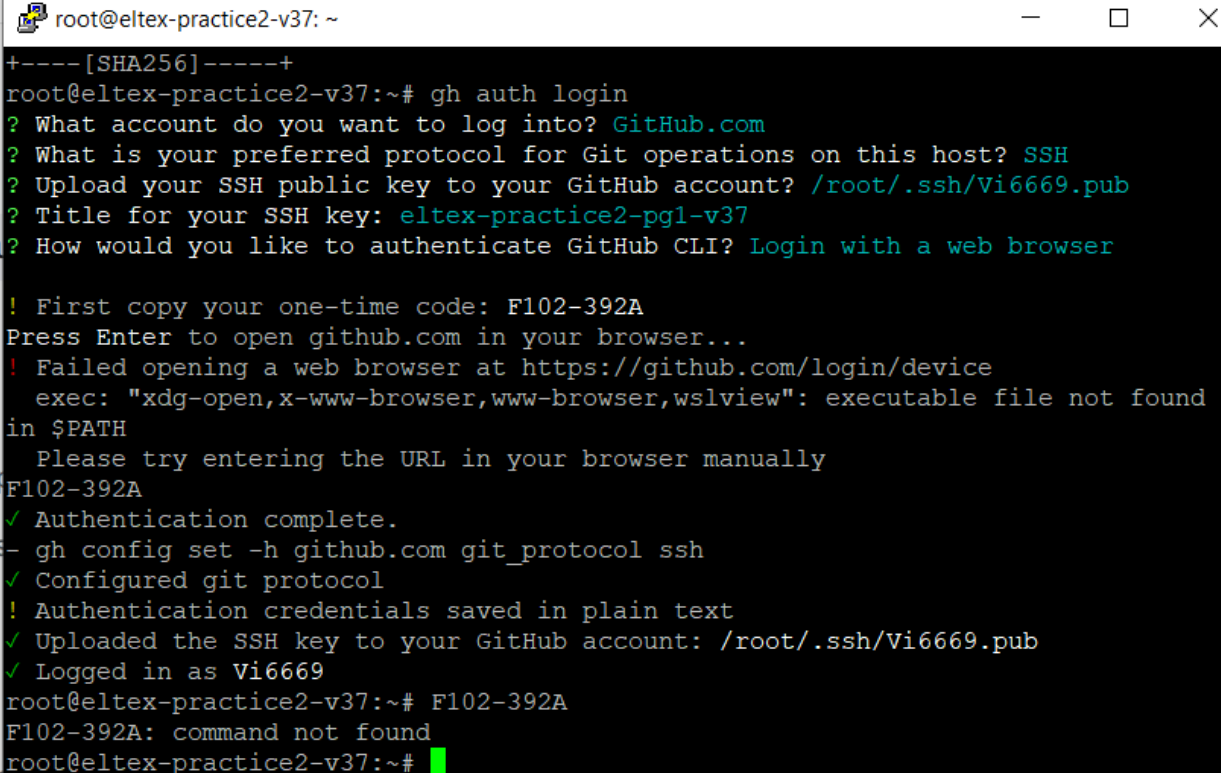
```
ssh-keygen -t ed25519 -f ~/.ssh/Vi6669
```



```
root@eltex-practice2-v37:~# ssh-keygen -t ed25519 -f ~/.ssh/Vi6669
Generating public/private ed25519 key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/Vi6669
Your public key has been saved in /root/.ssh/Vi6669.pub
The key fingerprint is:
SHA256:jpZ7BDtNO4X74YP4Vmb7JuCmITYirfAwhxBxLLTJbaw root@eltex-practice2-v37
The key's randomart image is:
+--[ED25519 256]--+
|oo.               |
|oo*              |
|. = +            |
| .o   . o .      |
|.E      =S+       |
|.o   o+B =        |
|* + + ==.X o     |
|.B o +.o* * .    |
|. .   o* . =.     |
+----[SHA256]-----+
root@eltex-practice2-v37:~#
```

Рисунок 29 – Ключ

gh auth login



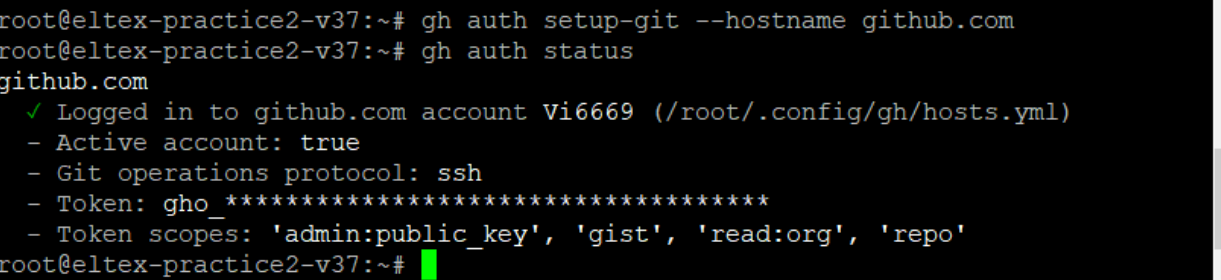
```
root@eltex-practice2-v37: ~
+----[SHA256]-----+
root@eltex-practice2-v37:~# gh auth login
? What account do you want to log into? GitHub.com
? What is your preferred protocol for Git operations on this host? SSH
? Upload your SSH public key to your GitHub account? /root/.ssh/Vi6669.pub
? Title for your SSH key: eltex-practice2-pgl-v37
? How would you like to authenticate GitHub CLI? Login with a web browser

! First copy your one-time code: F102-392A
Press Enter to open github.com in your browser...
! Failed opening a web browser at https://github.com/login/device
exec: "xdg-open,x-www-browser,www-browser,wslview": executable file not found
in $PATH
Please try entering the URL in your browser manually
F102-392A
✓ Authentication complete.
- gh config set -h github.com git_protocol ssh
✓ Configured git protocol
! Authentication credentials saved in plain text
✓ Uploaded the SSH key to your GitHub account: /root/.ssh/Vi6669.pub
✓ Logged in as Vi6669
root@eltex-practice2-v37:~# F102-392A
F102-392A: command not found
root@eltex-practice2-v37:~#
```

Рисунок 30 – Ключ

gh auth setup-git --hostname github.com

gh auth status



```
root@eltex-practice2-v37:~# gh auth setup-git --hostname github.com
root@eltex-practice2-v37:~# gh auth status
github.com
✓ Logged in to github.com account Vi6669 (/root/.config/gh/hosts.yml)
- Active account: true
- Git operations protocol: ssh
- Token: gho_*****
- Token scopes: 'admin:public_key', 'gist', 'read:org', 'repo'
root@eltex-practice2-v37:~#
```

Рисунок 31 – Статус

gh repo create --public eltex-practice

mkdir ~/eltex-practice && cd ~/eltex-practice

echo "# Учебный репозиторий Eltex - Практика 2" >> README.md

git init

git add README.md

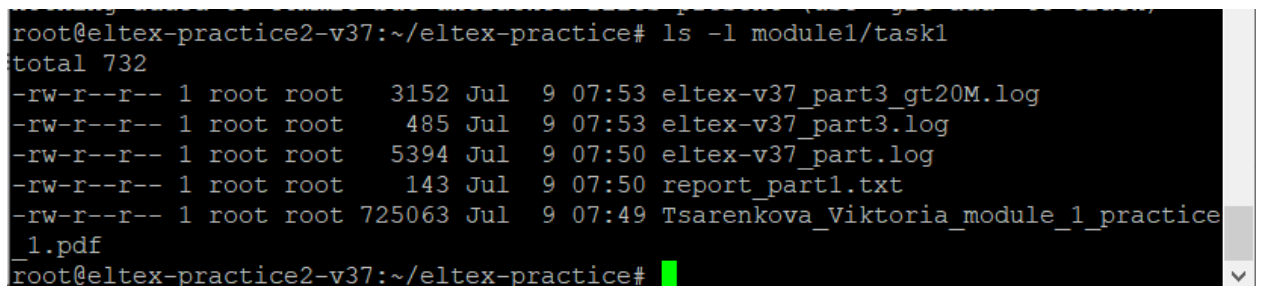
git commit -m "first commit"

git branch -M main

```

git remote add origin git@github.com:Vi6669/eltex-practice.git
git push -u origin main
echo -e "Host github.com\n  IdentityFile ~/.ssh/Vi6669" >> ~/.ssh/config
chmod 600 ~/.ssh/config
ssh -T git@github.com
git push -u origin main
mkdir -p module1/task1
mkdir -p module1/task2
mkdir -p module1/task3
exit
ls
scp -P 44556 "C:\Users\vikat\OneDrive\Рабочий стол\Tsarenkova_Viktoria_module_1_practice_1.pdf" eltex-v37@217.71.138.1:~/
scp ~/Tsarenkova_Viktoria_module_1_practice_1.pdf root@172.16.9.211:~/eltex-practice/module1/task1/
scp ~/eltex-v37_part.log root@172.16.9.211:~/eltex-practice/module1/task1/
scp ~/report_part1.txt root@172.16.9.211:~/eltex-practice/module1/task1/
scp ~/eltex-v37_part3.log root@172.16.9.211:~/eltex-practice/module1/task1/
scp ~/eltex-v37_part3_gt20M.log root@172.16.9.211:~/eltex-practice/module1/task1/
ssh root@172.16.9.211
ls -l module1/task1

```



```

root@eltex-practice2-v37:~/eltex-practice# ls -l module1/task1
total 732
-rw-r--r-- 1 root root 3152 Jul 9 07:53 eltex-v37_part3_gt20M.log
-rw-r--r-- 1 root root 485 Jul 9 07:53 eltex-v37_part3.log
-rw-r--r-- 1 root root 5394 Jul 9 07:50 eltex-v37_part.log
-rw-r--r-- 1 root root 143 Jul 9 07:50 report_part1.txt
-rw-r--r-- 1 root root 725063 Jul 9 07:49 Tsarenkova_Viktoria_module_1_practice_1.pdf
root@eltex-practice2-v37:~/eltex-practice#

```

Рисунок 32 – Файлы действительно скопировались

```
git add .
```

```
root@eltex-practice2-v37:~/eltex-practice# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   module1/task1/Tsarenkova_Viktoria_module_1_practice_1.pdf
    new file:   module1/task1/eltex-v37_part.log
    new file:   module1/task1/eltex-v37_part3.log
    new file:   module1/task1/eltex-v37_part3_gt20M.log
    new file:   module1/task1/report_part1.txt

root@eltex-practice2-v37:~/eltex-practice#
```

Рисунок 33 – Файлы действительно скопировались

git commit -m "add practice 1 files"

git push

mkdir ~/eltex-practice_clone && cd ~/eltex-practice_clone

git clone git@github.com:Vi6669/eltex-practice.git .

ls -la

cd ~/eltex-practice

echo "## Тестирование процессов Fetch и Merge" >> README.md

git add README.md

git commit -m "add test line for fetch-merge demo"

git push

cd ~/eltex-practice_clone

git fetch

git log origin/main

```

root@eltex-practice2-v37:~/eltex-practice_clone# git log origin/main
commit 2b25d234647616114493fd4977d453bc0e139041 (origin/main, origin/HEAD)
Author: Vi6669 <vika.tsarenkova@gmail.com>
Date: Thu Jul 9 08:11:43 2026 +0000

    add test line for fetch-merge demo

commit eed930c89f4af35c0c9e4cd23c3946365f56e64c (HEAD -> main)
Author: Vi6669 <vika.tsarenkova@gmail.com>
Date: Thu Jul 9 08:03:39 2026 +0000

    add practice 1 files

commit f8136cc0a1a3d2ac5341f0d8447b19b7943827b3
Author: Vi6669 <vika.tsarenkova@gmail.com>
Date: Thu Jul 9 07:16:16 2026 +0000

    first commit
root@eltex-practice2-v37:~/eltex-practice_clone#

```

Рисунок 34 – Изменения

git diff main origin/main

```

root@eltex-practice2-v37:~/eltex-practice_clone# git diff main origin/main
diff --git a/README.md b/README.md
index a178e59..398047e 100644
--- a/README.md
+++ b/README.md
@@ -1,2 @@
 # Учебный репозиторий Eltex - Практика 2
+## Тестирование процессов Fetch и Merge
root@eltex-practice2-v37:~/eltex-practice_clone#

```

Рисунок 35 – Разница

git merge